

Full length article

Mitigating systematic bias in polymer T_g prediction via role-decoupled graph readout[☆]Wanyi Chen^a, Zhanjie Liu^b, YiXuan Huo^a, Jun Liu^b, Lina Zhao^a, Lihong Cui^{a,*}^a College of Mathematics and Physics, Beijing University of Chemical Technology, 15 BeiSanhuan East Road, Beijing, 100029, China^b State Key Laboratory of Organic-Inorganic Composites, Beijing University of Chemical Technology, 15 BeiSanhuan East Road, Beijing, 100029, China

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ABSTRACT

The accurate prediction of the glass transition temperature (T_g) is crucial for polymer design. Standard Graph Neural Networks (GNNs) exhibit a systematic bias when applied to this task: their homogeneous graph readout mechanism fails to distinguish the distinct functional roles of backbone and side-chain atoms, leading to structural role mixing and a consequent overestimation error that amplifies in the high- T_g regime. To address this, we propose a novel role-decoupled graph readout paradigm. Our framework, termed BRAG, explicitly separates the learning paths for backbone and side-chain atoms based on molecular topology, and then models their competitive contribution to segmental mobility via a physics-informed absolute-difference operation. This design injects a structure-inductive bias directly into the representation learning process. Extensive experiments demonstrate that our method not only achieves comparable overall accuracy but, more distinctively, effectively rebalances the prediction error distribution. It reduces the high- T_g region error (up to 6.3% relative to the standard GNN baseline and mitigates the systematic bias prevalent in standard GNNs, $p < 0.05$) and substantially mitigates the high- T_g systematic bias that plagues standard GNNs. Ablation studies confirm the necessity of both role-decoupling and the differential interaction. This work highlights that for materials with intrinsic functional heterogeneity, architectural alignment with domain knowledge is a more effective lever for achieving model robustness and physical consistency than increasing model complexity.

1. Introduction

The glass transition temperature (T_g) of polymers is a critical parameter that demarcates the transition from a rubbery to a glassy state, where large-scale segmental motion ceases. It fundamentally defines the material's thermodynamic state, thereby dictating its processing windows and potential application scenarios. Therefore, the development of accurate and efficient predictive methods for T_g is crucial for accelerating the rational design and discovery of novel polymers [1,2]. But traditional prediction methods, such as group contribution techniques, struggle with complex polymer architectures, creating a critical need for data-driven approaches [3].

In recent years, machine learning plays an important role in polymer science [4,5]. Among these, Graph Neural Networks (GNNs) have achieved revolutionary progress in molecular property prediction due to their ability to learn representations directly from molecular topological structures (atoms as nodes, chemical bonds as edges). This end-to-end learning framework demonstrates notable advantages over traditional Quantitative Structure-Activity Relationship (QSAR) methods that rely on complex handcrafted features [6–10].

While Graph Neural Networks (GNNs) hold great promise for polymer informatics, their direct application to the prediction of structure-sensitive properties like T_g faces a fundamental challenge arising from its underlying architectural premise. Polymer molecules are not simple collections of homogeneous atoms; they possess distinct functional subdivisions—the backbone and the side chains [11]. The backbone forms a continuous, rigid skeleton whose segmental cooperative motion capability is the primary determinant of T_g . In contrast, side chains modulate chain mobility through steric hindrance, polarity, and free volume effects [1]. Standard GNNs employ permutation-invariant readout operations (e.g., global sum or mean pooling) to generate graph-level representations, which inherently assume that all atoms contribute equally and are interchangeable for the target property [12]. This homogeneity assumption fundamentally conflicts with the heterogeneous physicochemical roles of atoms in polymers, particularly the distinct functions of backbone versus side-chain atoms. We term this mismatch structural role mixing. This mixing obscures the key physical determinants of T_g , preventing the model from disentangling the distinct contributions of backbone stiffness and side-chain steric effects.

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* Corresponding author.

E-mail addresses: liujun@mail.buct.edu.cn (J. Liu), zhaoln@mail.buct.edu.cn (L. Zhao), cuilh@mail.buct.edu.cn (L. Cui).

This leads to a systematic prediction bias that is most pronounced in the high- T_g regime, as evidenced by a significant higher variance-normalized bias, i.e., error scaled by the intrinsic variability of each region, for high- T_g polymers relative to their low- T_g counterparts.

To overcome the structural role mixing issue, prevailing strategies — including attention mechanisms and contrastive learning — primarily focus on refining node-level representations or imposing spatial constraints, yet they fail to fundamentally redesign the graph-level readout mechanism. Alternatively, some approaches incorporate subgraph patterns or geometric priors, but they still lack a dedicated design to model the functional asymmetry and explicit coupling between distinct node subsets (e.g., backbone vs. side chains) within a unified molecular graph.

Therefore, we posit that the key to overcoming this bottleneck is to fundamentally redesign the graph-level representation generation paradigm, injecting a structure-inductive bias that reflects the underlying physics of polymers. To this end, we propose the Backbone-and-Side-chain Role-Aware Graph neural network (BRAG) framework. The core of BRAG is a “decoupling-then-interaction” paradigm: it first explicitly decouples the representation learning paths for backbone and side-chain nodes based on topological rules, and then applies a physics-inspired interaction function — computing the absolute difference between their aggregated representations — to model their competitive contribution to segmental mobility. This design directly circumvents the structural role mixing problem, enabling the model to capture the structural features most relevant to T_g . The main contributions of this work are threefold:

- **Problem Formulation:** We identify, formalize, and term the structural role mixing phenomenon, a fundamental representational flaw arising from the conflict between the homogeneous readout of standard GNNs and the intrinsic functional asymmetry of polymer backbones and side chains. We then establish a direct causal link between this bottleneck and the systematic error imbalance observed in high- T_g predictions.
- **Methodology:** We propose a novel decoupling-then-interaction paradigm and instantiate it in the BRAG (Backbone-and-side-chain Role-Aware Graph) framework. BRAG explicitly decouples the representation learning paths based on topological roles, then re-integrates them via a physics-inspired absolute-difference interaction. This design injects a physically grounded, structure-inductive bias into the graph readout mechanism, achieving architectural alignment with domain knowledge without introducing significant complexity.
- **Empirical Validation:** We demonstrate that the introduced structural bias mitigates the systematic over/under-prediction in the high- T_g region, a limitation that persists across baseline models. BRAG achieves superior the overall accuracy of leading baselines, achieves the lowest high- T_g error across multiple thresholds and reduces the high- T_g systematics bias. Ablation studies confirm that both role-aware decoupling and differential interaction are necessary for this gain.

The remainder of this paper is organized as follows. Section 2 reviews related work. We formally define the problem of structural role mixing and systematic error imbalance in Section 3. The BRAG framework is detailed in Section 4. Experimental results and analysis are presented in Section 5, followed by discussion and conclusion in Section 6.

In summary, this work demonstrates that for material systems with intrinsic functional partitions — such as the backbone and side chains in polymers — explicitly incorporating role-aware, structure-inductive bias at the architectural level offers an effective strategy to improve the stability and physical consistency of property prediction models. The proposed paradigm provides a principled blueprint for developing more reliable and interpretable machine learning models in polymer informatics and beyond.

2. Related work

2.1. Graph neural networks for polymer property prediction

Graph Neural Networks (GNNs) have emerged as a prevalent methodology for predicting polymer properties [13]. The canonical pipeline involves representing polymer repeat units as molecular graphs, learning atomic embeddings through message passing, and subsequently aggregating them via graph pooling to obtain a global representation for regression [14]. Recent advancements have primarily evolved along three trajectories: (1) designing more expressive GNN architectures (e.g., GIN captures graph isomorphisms, PNA handles diverse bond types) to better capture local chemical environments [15, 16]; (2) incorporating attention mechanisms to differentially weight atomic contributions [17,18]; and (3) integrating supplementary domain knowledge, such as quantum chemical descriptors and topological fingerprints [19]. Such as Wu et al. [20] incorporated periodicity priors for crystalline polymers, and Liu et al. [21] combined wavelet transforms with transformers. Notwithstanding these innovations, a fundamental premise remains largely unchallenged: the direct applicability of standard, permutation-invariant graph pooling operations to polymers—systems with intrinsically heterogeneous structural roles.

2.2. Limitations and adaptations of graph readout operations

GNNs rely on permutation-invariant functions (e.g., sum, mean, or max pooling) for graph-level readout, guaranteeing invariance to node ordering [12,22]. While crucial for generalizability, this design inherently imposes a homogeneity bias, treating all nodes as functionally equivalent. For systems like polymers, where atoms serve distinct roles (e.g., backbone rigidity vs. side-chain modulation), this bias leads to the structural role mixing previously identified, which obscures critical structure–property relationships and undermines prediction stability, especially in extreme property regimes (e.g., high- T_g). Recent efforts to move beyond simple pooling include hierarchical pooling methods, which build multi-scale representations via learned clustering. Zang et al. [23] learned multi-scale representations via node clustering, while Han et al. [24] applied it to ionic conductivity prediction. However, these clusters are data-driven and not guaranteed to align with physically meaningful, role-based partitions like the backbone-side chain dichotomy. Subgraph-aware GNNs constitute another direction, aiming to capture the importance of molecular substructures. FunQG decomposed graphs into quotient structures, and Zhang et al. [25] enriched MPNNs with functional-group embeddings [26]. Yet, they predominantly target predefined chemical functional groups, failing to address the continuous topological role of the backbone versus the discrete, appended nature of side chains—a partition defined by connectivity and function, not just local chemistry [27].

2.3. Incorporating inductive biases in materials machine learning

In materials machine learning, embedding domain knowledge as an inductive bias is a well-established paradigm for enhancing both model performance and interpretability [28,29]. Illustrative examples include equivariant neural networks, which encode the rotational and translational symmetries of 3D space to learn geometrically meaningful representations, and graph partitioning strategies for crystalline materials, which explicitly model periodic boundary conditions [30, 31]. Boiko et al. [32] infused stereoelectronic effects into graph representations. Brierley-Croft et al. [28] integrated group contributions, and Zhang et al. [10] designed polymer-unit fingerprints (PUFp) for organic semiconductors. Recent consistency-regularized GNNs improved robustness but remain role-agnostic [29]. These successes underscore the power of architecturally hardwiring physical or structural priors. Despite this progress, a conspicuous gap remains for polymer systems. Specifically, there has been little dedicated work on designing graph

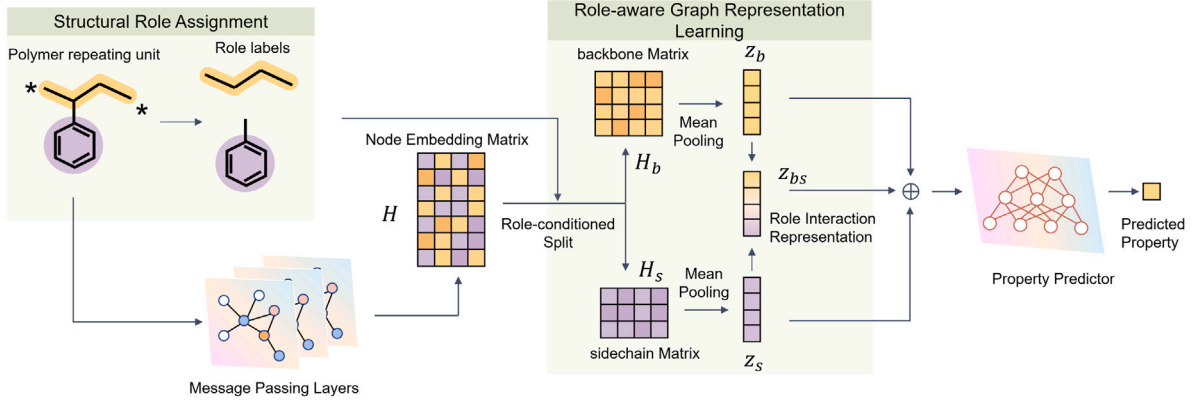


Fig. 1. Architecture of the role-aware graph representation learning framework for polymer glass transition temperature prediction. A polymer repeating unit is first converted into a molecular graph and processed through multiple message passing layers to obtain the node embedding matrix H . Based on a topological rule, nodes are then separated into backbone and side-chain components, forming role-specific feature matrices H_b and H_s . Role-specific mean pooling are applied to derive the corresponding role representations z_b and z_s . The structural interaction between the two roles is further captured by computing an interaction representation $z_{bs} = |z_b - z_s|$. These representations are concatenated into $[z_b \oplus z_s \oplus z_{bs}]$ and fed into an MLP for the final T_g prediction.

representation learning inductive biases that explicitly encode the ubiquitous and physically decisive backbone-side chain functional partition. The Backbone-and-Side-chain Role-Aware Graph (BRAG) framework proposed herein directly addresses this gap by introducing a role-aware, decoupling-then-interaction paradigm that injects this specific structural prior into the graph representation learning process.

3. Problem definition

3.1. Structural role mixing

Given a molecular graph of a polymer repeat unit $G = (V, E)$, the node set V can be partitioned into two mutually exclusive subsets according to their physicochemical roles: the backbone node set V_b and the side-chain node set V_s , such that

$$V = V_b \cup V_s, \quad V_b \cap V_s = \emptyset.$$

Backbone nodes constitute the molecular skeleton and dominate the cooperative segmental motion and overall rigidity of the polymer. In contrast, side-chain nodes primarily influence free volume and local intermolecular interactions. A conventional GNN obtains node representations $h_i \mid i \in V$ through message passing, and then applies a permutation-invariant readout function $R(\cdot)$ (e.g., global mean pooling) to produce a graph-level representation

$$h_G = R(h_i \mid i \in V),$$

which is subsequently used to predict the target property (e.g., T_g). Under symmetric permutation-invariant aggregation, this procedure implicitly assumes that

$$\forall i, j \in V, \quad \frac{\partial R}{\partial h_i} \approx \frac{\partial R}{\partial h_j},$$

i.e., each node contributes approximately homogeneously to the final graph representation.

However, for polymer T_g , the true physical determining function $f(G)$ is expected to treat nodes in V_b and V_s asymmetrically:

$$f(G) = g(\phi(h_i \mid i \in V_b), \psi(h_j \mid j \in V_s)),$$

where ϕ and ψ may be different aggregation functions.

The use of a single, undifferentiated aggregation function R in standard GNNs cannot capture such asymmetry. Consequently, the distinct functional information from backbone and side chains becomes mixed in the graph representation h_G , a phenomenon we term structural role mixing.

3.2. Systematic error imbalance

Structural role mixing hinders the model's ability to accurately capture the dominant contribution of the backbone to T_g , especially for polymers in the high- T_g regime where backbone rigidity is paramount. This representational shortcoming manifests empirically as a systematic error imbalance in prediction performance.

To quantitatively analyze prediction stability across different T_g intervals, we divide the dataset into high- T_g and low- T_g regions using a threshold τ . To ensure robustness, we evaluate multiple thresholds (e.g., 70th, 75th, 80th, and 90th percentiles); here we present results for the 75th percentile as a representative case. We then compute the mean absolute error (MAE) for each region:

$$\text{MAE}_{\text{high}}, \quad \text{MAE}_{\text{low}}.$$

We define the error imbalance metric Δ_{bias} as:

$$\Delta_{\text{bias}} = \text{MAE}_{\text{high}} - \text{MAE}_{\text{low}}.$$

When $\Delta_{\text{bias}} > 0$, the model exhibits larger prediction errors in the high- T_g region compared to the low- T_g region, indicating an error imbalance across T_g regimes.

To further quantify the imbalance while accounting for potential heteroscedasticity in the data, we introduce Normalized Bias. This metric normalizes regional errors by the corresponding standard deviation of T_g values in each region:

$$\text{Normalized Bias} = \frac{\text{MAE}_{\text{high}}}{\sigma_{\text{high}}} - \frac{\text{MAE}_{\text{low}}}{\sigma_{\text{low}}}.$$

Here, MAE_{high} and MAE_{low} denote the mean absolute errors in the high- T_g region ($T_g \geq 508.15\text{K}$) and low- T_g region ($T_g < 508.15\text{K}$), respectively; σ_{high} and σ_{low} represent the standard deviations of T_g values within the corresponding regions.

By variance-normalizing the regional errors, this metric measures the relative error intensity rather than absolute error differences across property intervals. For an ideal model, the Normalized Bias should be close to zero, indicating balanced predictive capability across different T_g regimes.

While Δ_{bias} quantifies the absolute error disparity between regions, which can be influenced by the inherent spread of T_g values, the Normalized Bias measures the relative error intensity, providing a scale-invariant assessment that is more robust to heteroscedasticity across the T_g range.

4. Methods

To address the structural role mixing problem defined in Section 3.1, we propose the Backbone-and-Side-chain Role-Aware Graph neural network (BRAG) framework. The core design principle is to explicitly decouple and then interact the representations of nodes with distinct functional roles at the graph-level representation stage, thereby injecting the necessary inductive bias directly into the readout mechanism rather than through modifications to the node encoding stage.

BRAG consists of three key steps: role separation, role-specific pooling, and constrained differential interaction. The overall architecture of BRAG is illustrated in Fig. 1.

4.1. Shared node encoding

Given a polymer molecular graph G and its initial node features X , we first apply a shared GNN encoder (e.g., a GCN) to learn intermediate representations for all nodes:

$$H = \text{GNN}(X, E) = h_1, h_2, \dots, h_n, \quad h_i \in \mathbb{R}^d.$$

This step is identical to standard GNN pipelines and aims to capture fundamental atomic features and local chemical environment information, providing a common foundation for the subsequent role-specific processing.

4.2. Structural role separation

Based on the topology of the polymer repeat unit, we apply a deterministic rule to partition the nodes into backbone node set V_b and side-chain node set V_s using a rule grounded in chemical prior knowledge: backbone nodes are defined as all atoms located on the shortest path between the wildcard markers ([*]) that connect repeat units; all remaining atoms are assigned to V_s . This partition is rule-based and requires no learnable parameters.

We then separate the corresponding node representations:

$$H_b = h_i \mid i \in V_b, \quad H_s = h_j \mid j \in V_s.$$

4.3. Role-specific pooling

To aggregate information from each role-specific subgraph while mitigating the influence of molecular size, we apply mean pooling independently to H_b and H_s , obtaining role-specific graph-level vectors:

$$z_b = \frac{1}{|V_b|} \sum_{i \in V_b} h_i, \quad z_s = \frac{1}{|V_s|} \sum_{j \in V_s} h_j.$$

Mean pooling is chosen because it provides a statistical expectation of the subgraph features for each role and remains robust to variations in node counts.

4.4. Differential interaction

The glass transition temperature T_g fundamentally reflects the freezing of segmental motion, which depends on the competition between rigidity constraints provided by the backbone and free volume introduced by the side chains.

To model this physical mechanism, we compute the element-wise absolute difference between z_b and z_s :

$$z_{bs} = |z_b - z_s|.$$

The absolute difference operation eliminates sign ambiguity and compels the model to focus on the magnitude of the disparity between the two structural contributions. Consequently, z_{bs} can be interpreted as a measure of the mismatch between backbone and side-chain constraint strength.

Table 1

Performance comparison of BRAG and baseline models. **MAE**: Mean Absolute Error; **RMSE**: Root Mean Square Error; **R²**: Coefficient of Determination; **High- T_g MAE**: metric computed on the high- T_g region (top 25% of samples, threshold 508.1 K fixed as the 75th percentile of the full dataset). All values are mean $\pm$ std over 10 random seeds with a shared data split (seed 2024). Superscripts report the most significant pairwise two-sided paired t -test between BRAG and the three primary baselines (BaseGNN, AttnGNN, Contrastive); significance levels: * $p < 0.05$, ** $p < 0.01$. Full pairwise statistics with 95% confidence intervals and effect sizes are reported in Section 5.3.1.

Model	MAE (K)	RMSE (K)	R ²	High- T_g MAE (K)
BaseGNN	25.76 $\pm$ 0.49	36.99 $\pm$ 0.75	0.896 $\pm$ 0.004	26.20 $\pm$ 1.19
AttnGNN	25.35 $\pm$ 0.42	36.97 $\pm$ 0.72	0.896 $\pm$ 0.004	25.83 $\pm$ 0.87
Contrastive	25.58 $\pm$ 0.32	36.53 $\pm$ 0.35	0.898 $\pm$ 0.002	26.11 $\pm$ 0.79
BRAG	24.77 $\pm$ 0.47**	35.62 $\pm$ 0.71**	0.903 $\pm$ 0.004**	24.54 $\pm$ 1.15**
<i>2 $\times$ 2 comparison over GNN encoder type and the structural readout bias</i>				
VanillaGIN	28.27 $\pm$ 1.48	39.60 $\pm$ 1.37	0.880 $\pm$ 0.009	29.98 $\pm$ 3.22
VanillaGAT	24.83 $\pm$ 0.92	38.10 $\pm$ 1.11	0.889 $\pm$ 0.006	25.52 $\pm$ 1.20
BRAG+GIN	26.29 $\pm$ 1.39	37.95 $\pm$ 1.58	0.890 $\pm$ 0.009	27.44 $\pm$ 2.02
BRAG+GAT	23.70 $\pm$ 0.80	36.22 $\pm$ 1.47	0.900 $\pm$ 0.008	25.16 $\pm$ 1.62

A larger z_{bs} implies a greater structural mismatch between backbone rigidity and side-chain flexibility, which the model learns to associate with increased segmental constraints and potentially higher T_g . This design provides explicit physical interpretability.

We demonstrate the superiority of this interaction function over alternatives (e.g., concatenation, weighted averaging) in the ablation study (Section 5.4).

4.5. Prediction layer

The final prediction is obtained by a lightweight multilayer perceptron (MLP) that takes the concatenated role and interaction representations as input:

$$\hat{y} = \text{MLP}(z_b \oplus z_s \oplus z_{bs}).$$

Since the decisive structural information is already encoded in z_{bs} , only a shallow MLP is required, ensuring model efficiency.

5. Results

5.1. Experimental setup

Dataset: We use the Openpoly polymer T_g dataset, a standard benchmark comprising 8469 unique polymer repeat units represented by PSMILES strings alongside their experimental T_g values [33]. The T_g values range from 130.0 K to 975.2 K, with a mean of 416.6 K and a standard deviation of 115.5 K. To analyze systematic bias, we partition the dataset into high- T_g and low- T_g regimes using a threshold $\tau = 508.15$ K, corresponding to the 75th percentile of the distribution; samples with $T_g \geq \tau$ constitute 25.3% of the data. On average, each polymer contains 35.3 atoms. Notably, the number of backbone and side-chain atoms is approximately equal, providing a balanced basis for evaluating role separation.

Evaluation Metrics: We evaluate the overall performance using Mean Absolute Error (MAE) and the coefficient of determination R^2 . To quantify systematic bias discussed in Section 3.2, we employ the error imbalance metric Δ_{bias} and the Normalized Bias.

- A positive Δ_{bias} indicates larger absolute errors in the high- T_g region.
- The Normalized Bias provides a scale-invariant assessment of relative error intensity.

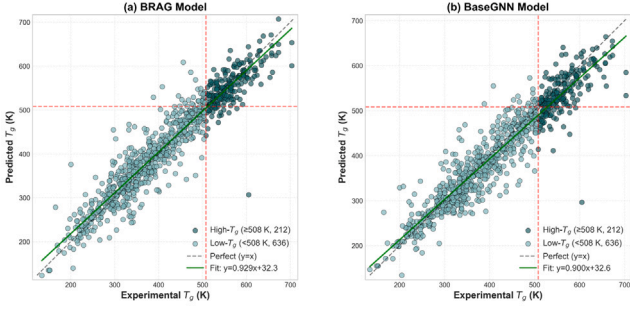


Fig. 2. Scatter plots comparing the glass transition temperature (T_g) prediction accuracy of BRAG and BaseGNN models on the polymer dataset. Each point represents a single polymers, with colors distinguishing high- T_g (≥ 508 K) and low- T_g (< 508 K) polymers. The diagonal dashed line represents perfect prediction ($y = x$).

All experiments are conducted using a fixed random seed with 10 repetitions, and we report the mean and standard deviation.

Baseline Models: We compare BRAG against three GNN baselines: (1) *BaseGNN*: a standard 3-layer GCN model serving as the vanilla reference without any role information; (2) *AttnGNN*: a GCN variant that learns node-level attention weights to adaptively re-weight atoms during the readout stage; (3) *Contrastive*: a GCN model jointly trained with a role-level contrastive objective that separates backbone and side-chain representations as an auxiliary loss. For a fair comparison, the three baselines and BRAG share an identical GCN encoder backbone with a hidden dimension of 128 and identical training hyperparameters (learning rate, batch size, etc.). In addition, to disentangle the contribution of the structural bias from the capacity of the underlying GNN encoder, we report two pure encoder baselines (*VanillaGIN*, *VanillaGAT*) together with BRAG variants built on the same alternative encoders (*BRAG+GIN*, *BRAG+GAT*) in Table 1, forming a 2×2 comparison over the encoder type and the structural bias.

BRAG Implementation: The structural role separation is performed via deterministic parsing of the PSMILES strings. The interaction representation is computed as the element-wise absolute difference $z_{bs} = |z_b - z_s|$. The final prediction is made by a two-layer MLP that takes the concatenated vector $[z_b \oplus z_s \oplus z_{bs}]$ as input.

5.2. Overall performance comparison

The overall predictive performance of BRAG and baseline models is quantitatively summarized in Table 1 and visually synthesized in Fig. 4 (radar chart). Among the GCN-based methods, BRAG achieves the lowest overall MAE (24.77 ± 0.47 K), corresponding to relative improvements of 3.8%, 2.3%, and 3.2% over BaseGNN, AttnGNN, and Contrastive, respectively. Pairwise two-sided paired t -tests over ten independent runs (Section 5.3.1) confirm that the improvements in MAE, RMSE, and R^2 are statistically significant relative to all three baselines (all $p < 0.05$; Cohen's $d > 0.9$ for MAE). While these overall accuracy gains are meaningful, the most pronounced advantage of BRAG emerges in the high- T_g regime, where the structural bias is most severe.

Fig. 3 shows that BRAG's advantage is concentrated in the high- T_g regime. It attains a high- T_g MAE of 24.54 ± 1.15 K, which is 5.0%, 6.3%, and 6.0% lower than AttnGNN, BaseGNN, and Contrastive, respectively. Pairwise paired t -tests confirm that this reduction is statistically significant relative to all three baselines (all $p < 0.05$; Section 5.3.1). The radar chart (Fig. 4) further consolidates this finding, showing BRAG's superior or comparable performance across all five evaluation metrics, with strengths in High- T_g MAE and Normalized Bias. This indicates that the role-aware design of BRAG yields larger benefits in the high- T_g region, which is theoretically more affected by the structural role mixing issue.

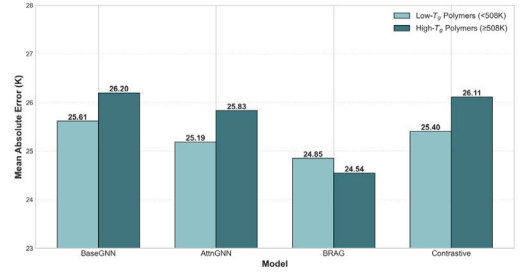


Fig. 3. Bar chart comparing the mean absolute error (MAE) of different graph neural network models on low- T_g (< 508 K) and high- T_g (≥ 508 K) polymers.

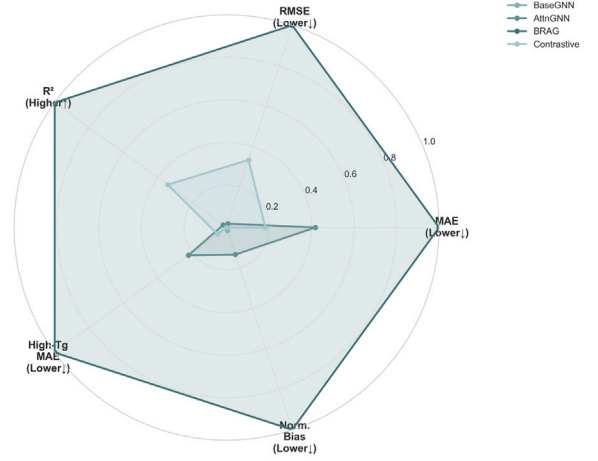


Fig. 4. Radar chart visualizing the normalized performance of BRAG and baseline models across five evaluation metrics: MAE, RMSE, R^2 , High- T_g MAE, and Normalized Bias. All metrics are normalized to $[0, 1]$ scale where higher values indicate better performance.

Table 2

Error distribution structure analysis. **MAE (All)**: Mean Absolute Error over all samples; **Normalized Bias**: bias normalized by standard deviation; **High- T_g MAE**: MAE for high glass transition temperature samples; **Δ Bias**: MAE difference between high- T_g and low- T_g regions.

Model	MAE (All) (K)	Normalized bias	High- T_g MAE (K)	Δ Bias (K)
BaseGNN	25.76 ± 0.49	$+0.318 \pm 0.037$	26.20 ± 1.19	$+0.58 \pm 1.98$
AttnGNN	25.35 ± 0.42	$+0.315 \pm 0.024$	25.83 ± 0.87	$+0.64 \pm 1.22$
BRAG	24.77 ± 0.47	$+0.288 \pm 0.030$	24.54 ± 1.15	-0.30 ± 1.47
Contrastive	25.58 ± 0.32	$+0.319 \pm 0.021$	26.11 ± 0.79	$+0.71 \pm 1.05$

5.3. Analysis of error distribution and systematic bias

To quantitatively assess whether BRAG mitigates the systematic error imbalance defined in Section 3.2, we analyze the detailed error structure presented in Table 2 and visualized in Fig. 5.

BRAG maintains a leading overall MAE of 24.77 ± 0.47 K across 10 independent runs. As shown in Fig. 5, it consistently rebalances the error distribution. This is evidenced by two key metrics:

- **Normalized Bias:** BRAG achieves the lowest value ($+0.288 \pm 0.030$), lower than all three baselines (BaseGNN $+0.318 \pm 0.037$, AttnGNN $+0.315 \pm 0.024$, Contrastive $+0.319 \pm 0.021$). This reduction is statistically significant relative to the contrastive learning model, while the differences relative to BaseGNN and AttnGNN do not reach the $p < 0.05$ level. These results confirm BRAG's capability in equilibrating the relative error intensity between high- T_g and low- T_g regions, with the most pronounced

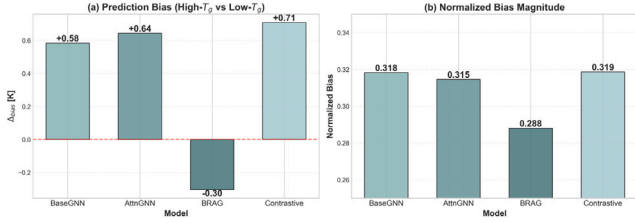


Fig. 5. Bar charts analyzing prediction bias across different models. (a) $\Delta_{bias} = MAE_{high} - MAE_{low}$, where positive values indicate worse performance on high- T_g polymers. (b) Normalized Bias = $\frac{MAE_{high}}{\sigma_{high}} - \frac{MAE_{low}}{\sigma_{low}}$, accounting for heteroscedastic noise differences between high- T_g and low- T_g data distributions.

improvement observed when compared to the contrastive baseline.

- Error Imbalance (Δ_{bias}): While baseline models exhibit a positive Δ_{bias} (e.g., $+0.58 \pm 1.98$ K for BaseGNN), indicating larger absolute errors for high- T_g samples, BRAG's Δ_{bias} shifts to -0.30 ± 1.47 K. The movement towards zero (Fig. 5) suggests that BRAG effectively counteracts the systematic bias prevalent in standard GNNs.

The scatter plot in Fig. 2 provides an intuitive visual confirmation of this bias correction. Compared to BaseGNN, whose predictions for high- T_g samples (orange points) show a clearer tendency to scatter above the parity line (indicating overestimation), BRAG's predictions are more symmetrically distributed around the ideal $y = x$ line across the entire T_g range. This visual evidence aligns perfectly with the quantitative metrics: BRAG's MAE in the high- T_g region (24.54 ± 1.15 K) is lower than BaseGNN and Contrastive, while the difference from AttnGNN is not statistically significant.

Collectively, these results validate our core hypothesis: by explicitly decoupling and interacting backbone-side chain representations, BRAG alleviates the structural role mixing problem. This leads to a more stable and reliable prediction model, as evidenced by the rectified error distribution—particularly the reduction of systematic overestimation in the high- T_g regime. The consistent improvements across all 10 runs confirm the robustness of these findings.

5.3.1. Statistical significance analysis

To rigorously assess the significance of the observed improvements, we conducted pairwise two-sided paired t -tests between BRAG and each of the three primary baselines (BaseGNN, AttnGNN, and Contrastive) over the 10 independent runs, all sharing the same data split (seed 2024); the full results are summarized in Table 3. For the overall MAE, BRAG is significantly better than all three baselines (all $p < 0.05$, Cohen's $d \geq 0.97$). The RMSE and R^2 improvements are likewise significant relative to all three baselines (all $p < 0.01$). In the high- T_g region, the MAE reduction is significant relative to all three baselines (all $p < 0.05$, Cohen's $d \geq 0.92$), and BRAG also achieves a significantly lower low- T_g MAE than Contrastive ($p = 0.024$), while remaining statistically on par with BaseGNN and AttnGNN. The reduction in normalized bias is significant relative to Contrastive ($p = 0.028$, Cohen's $d = 0.83$), but not relative to BaseGNN ($p = 0.095$) or AttnGNN ($p = 0.051$). The difference in Δ_{bias} does not reach significance against any baseline (all $p > 0.09$). We therefore conclude that BRAG delivers statistically significant improvements in overall accuracy (MAE, RMSE, and R^2) and in the high- T_g regime relative to all three primary baselines, complemented by a significant reduction in normalized bias relative to the contrastive baseline.

5.3.2. Multi-threshold error analysis

The main experiment defines the high- T_g regime using a fixed threshold (508.15 K, the 75th percentile of the full dataset). To verify that the reported bias mitigation is not an artifact of this particular threshold choice, we repeat the evaluation with the high- T_g boundary set to the 70th, 75th, 80th, and 90th percentiles of the test-set T_g distribution, keeping all other settings unchanged (Table 4). Following the main experiments, every model is trained over the same ten independent seeds, and the multi-threshold metrics are computed on the seed-averaged (ensemble) predictions; they are therefore reported as single values without standard deviations. At the 70th and 75th percentiles, BRAG achieves the lowest high- T_g MAE (21.98 and 22.40 K, respectively). At the stricter 80th and 90th percentiles, where the high- T_g subset shrinks to 20% and 10% of the samples and all models degrade, BRAG remains comparable to the best baseline (23.50 vs. 23.46 K at the 80th percentile and 27.21 vs. 27.17 K at the 90th percentile, both for AttnGNN), while clearly outperforming BaseGNN and Contrastive at every threshold (e.g., 27.21 vs. 28.02 and 28.72 K at the 90th percentile). The normalized bias of BRAG is likewise the lowest at the 70th and 75th percentiles (0.207 and 0.277) and within 0.003 of AttnGNN at the remaining thresholds. The same conclusion holds for the GIN-based variants: BRAG+GIN (22.97, 23.42, 23.96, and 27.38 K) outperforms VanillaGIN (27.91, 28.54, 30.11, and 36.49 K) by a large margin at every threshold, mirroring the GCN- and GAT-based comparisons. These results demonstrate that the bias mitigation of BRAG persists across a wide range of threshold choices rather than being tuned to a single operating point.

5.4. Ablation and mechanism analysis

To dissect the contribution of each design choice in BRAG and validate its underlying mechanisms, we conduct systematic ablation studies. The results, summarized in Table 5 and Table 6 with the supporting visualization in Fig. 6, provide evidence for our architectural decisions.

5.4.1. Necessity of role-aware design

The ablation studies first verify that the performance gain stems from the physically correct role separation, not from increased model capacity or stochastic factors.

- BRAG (shuffled): Randomly reassigning backbone and side-chain roles while keeping the model architecture identical causes a 17.7% increase in MAE (29.34 vs. 24.92 K), rendering performance worse than the BaseGNN baseline. This demonstrates that the gain is attributable to the correct, topology-based role partition.
- w/o backbone, w/o side-chain: Removing either the backbone or the side-chain pooling path (i.e., reverting to a single-path model) leads to significant performance degradation (MAE increases by 2.6% and 3.5%, respectively). Crucially, the error imbalance metric Δ_{bias} for these variants reverts to a large positive value, similar to the baseline models. This confirms that both structural roles are indispensable and must be processed through separate pathways to prevent the information mixing that causes systematic bias.

5.4.2. Analysis of interaction functions

We further investigate the optimal way to integrate the decoupled role representations. As shown in Table 6 and vividly compared in the heatmap (Fig. 6), the choice of interaction function is critical.

The simple concatenation of z_b and $z_s(\text{cat})$ yields suboptimal results. The direct difference (diff) performs better and the absolute difference (abs_diff) achieves the highest R^2 together with a near-optimal high- T_g MAE, on par with the best competing interaction. This finding aligns

Table 3

Pairwise two-sided paired t -tests between BRAG and each baseline over the 10 independent runs (shared split, seed 2024). Each cell reports the p -value with significance stars and the absolute Cohen's d effect size (in parentheses); n.s. denotes $p \geq 0.05$. Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Metric	BaseGNN	AttnGNN	Contrastive
MAE (K)	$p = 0.0013^{**}$ ($d = 1.46$)	$p = 0.0133^*$ ($d = 0.97$)	$p = 0.0013^{**}$ ($d = 1.45$)
RMSE (K)	$p = 0.0029^{**}$ ($d = 1.28$)	$p = 0.0046^{**}$ ($d = 1.19$)	$p = 0.0052^{**}$ ($d = 1.16$)
R^2	$p = 0.0029^{**}$ ($d = 1.28$)	$p = 0.0046^{**}$ ($d = 1.18$)	$p = 0.0051^{**}$ ($d = 1.16$)
High- T_g MAE (K)	$p = 0.0177^*$ ($d = 0.92$)	$p = 0.0145^*$ ($d = 0.95$)	$p = 0.0082^{**}$ ($d = 1.07$)
Low- T_g MAE (K)	$p = 0.0534$ (n.s.)	$p = 0.2415$ (n.s.)	$p = 0.0240^*$ ($d = 0.86$)
Normalized bias	$p = 0.0955$ (n.s.)	$p = 0.0506$ (n.s.)	$p = 0.0279^*$ ($d = 0.83$)
Δ_{bias} (K)	$p = 0.3039$ (n.s.)	$p = 0.1411$ (n.s.)	$p = 0.0928$ (n.s.)

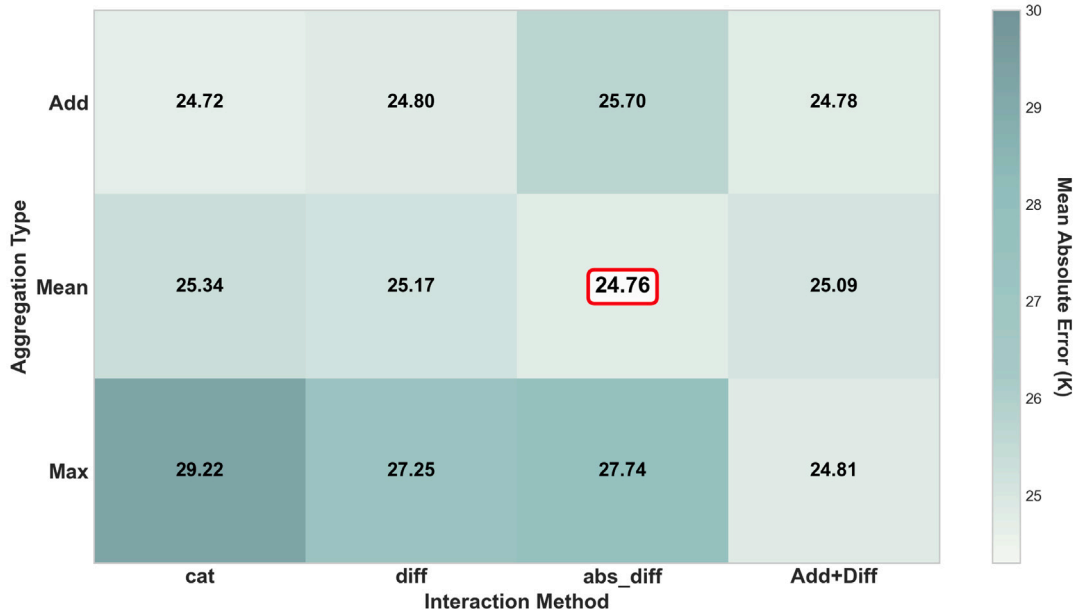


Fig. 6. Heatmap displaying the mean absolute error (MAE) of different aggregation strategies and interaction methods for polymer graph representation learning. Color intensity indicates performance (lighter = better), with numerical values showing MAE in Kelvin. The red box marks the default configuration (mean pooling + abs_diff), selected by the highest R^2 (0.904 ± 0.003) rather than the lowest overall MAE.

Table 4

High- T_g error analysis across alternative thresholds. The high- T_g boundary is defined as the 70th, 75th, 80th, and 90th percentiles of the test-set T_g distribution. Values are the high- T_g MAE (K) computed on the seed-averaged predictions over ten independent runs.

Model	70th pct.	75th pct.	80th pct.	90th pct.
BaseGNN	23.21	23.65	24.56	28.02
AttnGNN	22.54	22.94	23.46	27.17
BRAG	21.98	22.40	23.50	27.21
Contrastive	23.14	23.35	24.33	28.72
VanillaGIN	27.91	28.54	30.11	36.49
BRAG+GIN	22.97	23.42	23.96	27.38
VanillaGAT	21.52	22.67	23.88	26.51
BRAG+GAT	20.08	20.51	21.59	23.90

Table 5

Ablation study of the BRAG framework. **MAE**: Mean Absolute Error; **R^2** : Coefficient of Determination; **High- T_g MAE**: MAE for high glass transition temperature samples.

Model variant	MAE (K)	R^2	High- T_g MAE (K)
BRAG-full	24.92 ± 0.55	0.902 ± 0.005	24.44 ± 1.21
BRAG (shuffled)	29.34 ± 0.96	0.873 ± 0.008	30.33 ± 1.12
BRAG (random)	31.38 ± 0.76	0.862 ± 0.007	33.04 ± 1.77
w/o backbone	25.58 ± 0.34	0.886 ± 0.004	26.50 ± 0.93
w/o side-chain	25.78 ± 0.70	0.892 ± 0.006	26.55 ± 0.91

Table 6

Pooling and interaction function analysis. Pooling: pooling method used; Interaction: interaction function between backbone and side-chain; R^2 : Coefficient of Determination; High- T_g MAE: MAE for high glass transition temperature samples.

Pooling	Interaction	R^2	High- T_g MAE (K)
Mean	cat	0.897 ± 0.004	24.95 ± 1.39
Mean	diff	0.898 ± 0.004	24.88 ± 1.17
Mean	abs_diff	0.904 ± 0.003	25.13 ± 0.96
Mean	add+diff	0.897 ± 0.004	26.20 ± 1.33
Add	cat	0.901 ± 0.003	25.59 ± 1.22
Add	diff	0.898 ± 0.003	25.86 ± 0.76
Add	abs_diff	0.891 ± 0.005	26.60 ± 0.97
Add	add+diff	0.899 ± 0.005	26.00 ± 1.05
Max	cat	0.807 ± 0.025	32.40 ± 2.23
Max	diff	0.813 ± 0.003	30.79 ± 1.02
Max	abs_diff	0.810 ± 0.003	30.92 ± 1.10
Max	add+diff	0.899 ± 0.003	24.96 ± 0.50

with the physical picture of T_g : it is determined by the magnitude of the mismatch between backbone rigidity and side-chain flexibility, rather than the signed difference or mere co-existence of features. The absolute difference operation inherently captures this competition-based disparity, providing an effective and interpretable inductive bias for the subsequent prediction layer. Consequently, mean pooling + abs_diff is established as the default configuration.

In summary, the ablation studies offer three-fold validation:

- The topology-guided, role-aware partitioning is the cornerstone of BRAG's effectiveness.
- The dual-pathway design is necessary to preserve and utilize the distinct information from each role.
- The absolute-difference interaction is the most effective mechanism to model the physics-relevant competition between the two roles.

These results mechanistically explain how BRAG mitigates structural role mixing, moving beyond correlation to establish a causal understanding of the framework's success.

5.5. Discussion

Our work identifies and addresses a fundamental representational bottleneck in applying GNNs to polymer T_g prediction: the systematic error imbalance in high- T_g predictions stems not from inadequate node feature learning, but from the homogeneity bias inherent in standard graph readout operations, which fails to respect the functional asymmetry between polymer backbones and side chains. BRAG addresses this by architecturally hardwiring a role partition into the readout through a simple “decouple-then-interact” paradigm, without increasing model complexity. The ablation studies confirm that both the correctness of the role partition and the differential interaction are indispensable, and the resulting representation directly encodes the physically meaningful competition between structural components, leading to a more interpretable model.

Consequently, BRAG's primary gain lies in reshaping the error landscape alongside statistically improvements in overall accuracy. This combination of enhanced average performance and bias mitigation is particularly valuable for materials discovery, where prediction reliability across the full T_g range matters more than isolated accuracy gains. Moreover, this is achieved with negligible computational overhead, since the role separation is rule-based and the pooling is linear. More broadly, for systems with inherent and well-defined structural or functional partitions, explicit architectural alignment with domain priors can be a more powerful strategy than merely pursuing greater network depth or width, suggesting potential applicability to other role-differentiated material systems, such as core/shell nanoparticles, composite interfaces, or multi-phase alloys.

Robustness of the role-separation rule. The backbone/side-chain partition is obtained by a deterministic rule: the backbone consists of the atoms on the shortest path between the two “*” wildcard atoms of the repeat unit in PSMILES, and all remaining atoms form the side chain; when fewer than two wildcard atoms are present, the entire unit is treated as backbone. Because the rule depends only on repeat-unit topology, it is well defined for the linear and linear-branched architectures that dominate the dataset and degrades gracefully on more complex topologies: for branched units, the shortest path selects one continuous chain direction with the remaining branches assigned to the side chain, and for ring-containing backbones, the ring atoms not on the path are treated as side chain—an approximation that is exact for short rigid linkers but may under-separate heavily cyclic backbones. For example, the partition is exact for polystyrene (*CC(*)c1ccccc1*), where the phenyl ring is correctly assigned to the side chain, whereas the fused carbazole ring of poly(vinyl carbazole) (*CC(*)c1cccc2c1[nH]c1ccccc12*) is treated as side chain only approximately. Irregular units that cannot be represented with two wildcard atoms fall back to a single-path readout, losing the bias gain without harming accuracy. To empirically validate that this fallback handling introduces no systematic bias, we compared keeping versus removing homopolymer samples (which contain no side chain, 1.43% of the dataset): excluding them changed the test MAE by only +0.74 K relative to the keep-all configurations. The ring and fallback cases nonetheless represent the main limitations of the current rule, and a learned-but-constrained role assignment is a promising direction for extending BRAG to arbitrary polymer topologies.

Orthogonality of the structural bias to the GNN encoder. A 2×2 comparison over encoder type (GCN, GIN, GAT) and the presence of the structural readout bias (Table 1) shows that BRAG consistently improves over each pure encoder (reducing the MAE of VanillaGIN from 28.27 ± 1.48 K to 26.29 ± 1.39 K and of VanillaGAT from 24.83 ± 0.92 K to 23.70 ± 0.80 K), demonstrating that the role-aware readout is orthogonal to, and composable with, the message-passing backbone. The encoder itself still contributes substantially—BRAG+GAT (23.70 K) outperforms BRAG with the GCN encoder (24.77 K)—so we recommend combining the BRAG readout with a strong encoder. A systematic comparison with further state-of-the-art architectures (e.g., PNA and transformer-based molecular graph models) under identical data splits and training protocols is left for future work, since the present study focuses on the structural readout bias rather than on maximizing absolute accuracy.

Future work can proceed in several directions: (1) extending the role-aware paradigm to other role-sensitive polymer properties (e.g., modulus, dielectric constant); (2) exploring learnable yet constrained role separation for systems where the partition is less unambiguous; (3) benchmarking against further state-of-the-art graph architectures under identical protocols; and (4) integrating this architectural bias with 3D geometric or dynamical information for a more holistic material representation.

6. Conclusion

In summary, this work identifies and addresses a fundamental representational flaw, structural role mixing, in applying graph neural networks to polymer T_g prediction. This flaw stems from a core mismatch: the homogeneous readout of standard GNNs is intrinsically misaligned with the heterogeneous physicochemical roles of backbone versus side-chain atoms.

To rectify this, we introduce the BRAG framework, which implements a decoupling-then-interaction paradigm. BRAG explicitly decouples atom representations based on topological roles and then reintegrates them via a physics-informed absolute-difference operation. This architectural design effectively injects a role-aware, structural inductive bias into the graph readout mechanism, thereby directly addressing the identified representational bottleneck.

Comprehensive experiments confirm that BRAG not only outperforms leading baselines in overall accuracy but, more importantly, consistently improves the balance of the error distribution. It significantly reduces prediction error in the challenging high- T_g regime and effectively mitigates the high- T_g systematic overestimation bias prevalent in standard GNNs, with paired statistical tests and multi-threshold analyses confirming that these gains are robust. Ablation studies mechanistically validate that both correct role separation and the differential interaction are indispensable.

Our findings highlight a key principle for reliable machine learning in materials science: for systems with well-defined functional partitions, architectural alignment with domain-specific structural knowledge offers a more effective and reliable strategy for achieving robustness than merely increasing model complexity. The BRAG framework provides a generalizable blueprint for building interpretable and stable predictive models, with clear potential for extension to other role-sensitive material systems, such as composites or multi-phase materials.

CRedit authorship contribution statement

Wanyi Chen: Writing – original draft, Methodology, Conceptualization. **Zhanjie Liu:** Writing – review & editing. **YiXuan Huo:** Writing – review & editing. **Jun Liu:** Funding acquisition. **Lina Zhao:** Funding acquisition. **Lihong Cui:** Supervision, Funding acquisition.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT in order to improve language clarity and polish the writing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The code and reference dataset [33] used in this study are listed below.

<https://github.com/cnqwyz/BRAG> (GitHub (our work))

[OpenPoly Database \(Reference data\)](#)

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